IBRAHIM LAWAL ABDULLAHI

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EXECUTIVE SUMMARY

An enthusiastic materials physicist highly proficient in the synthesis, structural optimization, and electrochemical characterization of advanced functional nanomaterials and have built a track record in peer-reviewed publications. Engaged in volunteering and outreach activities, committed to academic excellence. An awards-winning scholar and clean energy advocate dedicated to bridging experimental electrochemistry with data-driven solutions to accelerate the global transition toward resilient, sustainable power systems.

SKILLS AND ABILITIES

Laboratory: Electrochemical and materials characterization (CV, GCD, Cycling Stability, XRD, FESEM, FTIR, Raman Spectroscopy).
Technology/Software: Data Science, Python, MATLAB, Anaconda, OriginPro, and EC-Lab and Nova.
Programming Libraries/ Microsoft Office: NumPy, Pandas, SciPy, Matplotlib, Qiskit, Word, Excel (advanced), PowerPoint.
Soft Skills: Technical Manuscript Writing, Scientific Communication, Interdisciplinary Collaboration, and Critical Problem Solving.

EDUCATION

Master of Science in Physics [Condensed Matter Physics, Photonics]

Tezpur University (Central University) [2024 –2026]

Address: Sonitpur, Assam (India)

Final Grade: First Class | **Rank:** 2nd | **Relevant Courses:** Advanced CMP and Materials Science, Statistical Mechanics, Photonics.

Bachelor of Science in Physics

Umaru Musa Yar'adua University [2016 – 2020]

Address: Dutsin-Ma road, Katsina (Nigeria)

Final grade: Second Class Upper | **CGPA:** 4.31 out of 5.0 (Top: 3%) | **Relevant Courses:** Statistical Physics, Solid-State Physics.

RESEARCH AND TEACHING EXPERIENCE

Graduate Research Assistant

Electrochemistry Laboratory, Department of Physics, Tezpur University, India [2024 – 2026]

Research Topic: Structural and electrochemical investigations of Antimony Trioxide as an electrode material.

- Conducted extensive novel research on sustainable energy solutions and synthesized Sb_2O_3 as an electrode material.
- Executed rigorous material characterization (XRD, FESEM, FTIR, Raman Spectroscopy) and advanced electrochemical analysis (CV, GCD, Cycling Stability) to systematically evaluate phase kinetics in active architectures.

Supervisor: Dr. Shyamal Kumar Das

Achievement: Achieved A^+ grade in laboratory research; co-authored a manuscript (currently under review)

Professional Summer Research Intern

Advanced Functional Material Laboratory, Department of Physics, Tezpur University, India [2026]

Research Topic: Nanomaterial Synthesis and applications: Bi_2MoO_6 and Bi_2WO_6 for rechargeable aqueous Aluminum-ion battery.

- Synthesized bismuth-based metal oxide nanostructures via controlled hydrothermal routes to evaluate their viability as host frameworks for aqueous Aluminum-ion batteries.

Achievement: Presented research at *NanoX 2026* ("Bridging Theory and Applications"), detailing hydrothermal reaction kinetic

Summer Research Intern

Optical & Spectroscopy Laboratory, Department of Physics, Gauhati University, India [2025]

Research Topic: Study of Fluorescence Quenching Effects of (RhB) in Water and Propanol (Applied to Liquid-Phase Transport Phenomena)

- Leveraged high-precision optical and spectroscopic characterization techniques, building critical competencies transferable to using *in-situ* and *operando* optical diagnostic tools for monitoring battery cell degradation

Supervisor: Dr. Simanta Hazarika

Achievement: Received excellent feedback from the viva presentation committee

Machine Learning Research Intern

EngineerCore | IIIT Allahabad, India [2025]

◦ Applied supervised learning algorithms and time series analysis to model complex electrochemical data sets

Physics Teacher and Lab Instructor

Albarka International Schools, Katsina State [2019 – 2020]

◦ Taught Physics 1 to a class of 40 students and conducted laboratory demonstration such as verifying ohm's law amongst others.

PUBLICATIONS

Abdullahi, I. L., et al. "Hybrid sodium-ion battery anode systems: integrating advanced materials for high- performance storage" World Journal of Advanced Engineering Technology and Sciences.
DOI: https://doi.org/10.30574/wjaets.2025.16.3.1358

Abdullahi, I. L., et al. "Artificial Intelligence-driven control of lithium-ion battery systems for improved grid integration and power supply optimization" Global Journal of Engineering and Technology Advances.
DOI: https://doi.org/10.30574/gjeta.2025.24.3.0263

CONFERENCES AND WORKSHOP ATTENDED

Presenter: NanoX 2026 Program, Tezpur University, India. Presented "Nanomaterial Synthesis & Applications: Bi_2MoO_6 and Bi_2WO_6 for rechargeable aqueous Aluminum-ion battery. [2026]

Participant: International Conference on Sustainable Energy Solutions & Industry-Academia Meet, Tezpur University, India [2026]

◦ Fully engaged on lectures and panel discussions focused on industry needs, commercialization pathways, and collaborative research on innovative solutions for renewable energy, green hydrogen, energy storage, smart grids, and green materials.

Participant: 47th Annual Conference on Smart Physics, Nigerian Institute of Physics [2026]

◦ Engaged on lectures on innovative solutions for renewable energy, smart grids, next-generation energy storage, and advanced developments in green hydrogen production and infrastructure.

Participant: 2-Days Workshop on Advanced and Functional Materials for Energy Application, National Physics Laboratory, India [2026]

◦ Engaged in advanced sessions covering materials synthesis, characterization that focused on functional nanomaterial design and technologies for sustainable energy storage.

HONOURS AND AWARDS

Indian Council for Cultural Relations Scholarship (ICCR Scholarship): Fully funded scholarship for my MSc program. [2024-2026]
The International Physics Competition Certificate: Qualification round awardee. [2025]
Government of Nigeria Fellowship (3MTT Fellowship): Fully funded scholarship for my Data Science program. [2024]
Katsina State Government Merit Scholarship: Awarded for top academic performance in undergrad studies. [2016-2020]

VOLUNTEERING AND COMMUNITY SERVICE

Nigeria Youth Energy Ambassador Program: Led campaigns promoting clean energy technologies & sustainable energy storage. [2026]

Eco Ambassador, All-Africa Students Union: Developed capacity in green entrepreneurship and regional green economy acceleration. [26]

National Youth Service Corps (NYSC), Jigawa State, Nigeria: led grassroots climate awareness campaigns and advocated for localized sustainable energy solutions in rural northern Nigeria. [2021-2022]

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Digital Technology for Energy Efficiency: Asian Development Bank Institute [2026]

Next-Gen. Energy Storage Battery and Hydrogen Technology: Coursera [2026]

Introduction to Battery Management Systems: Coursera [2026]

Battery Technologies: Coursera [2026]

Data Science: National Information Technology Development Agency, Nigeria [2025]

◦ Preprocessed data using normalization and encoding techniques in Python and apply ML models to experimental datasets

Python4Physics: Remote Experience for Young Engineers and Scientists (REYES), University of California, Berkeley (UCB) [2025]

◦ Applied statistical mechanics energy formulations and Ising model spin configurations to simulate lattice-based phase transitions, establishing a theoretical foundation for modeling thermodynamic interactions and phase separation in battery electrode materials.

Data Analysis with Python-IBM Skills: Coursera [2023]

Renewable Energy Certification: Asian Development Bank Institute [2020]

PROFESSIONAL MEMBERSHIPS

Member, Nigerian Institute of Physics (NIP) [2026-Present]
Member, Society for Energy, Materials & Sustainability [2026-Present]
Member, National Association of Physics Students (NAPS) [2019-Present]
Member, Junior Engineers, Technicians & Scientists Club (JETS) [2014-2020]